Exercise : 21

Nitrogen bases.

Aim : To study and identify the structure of nitrogen bases

Principle: DNA and RNA are polymers of monomers called nucleotides, each of which is composed of a nitrogen base, a pentose sugar and a phosphate. The nitrogen base is a heterocyclic compound that can be either a purine (two rings) or a pyrimidine (one ring). There are 2 types of purines – adenine (A) and guanine (G) and 3 types of pyrimidines – cytosine (C), thymine (T) and uracil (U).

Requirements: Model / Photograph / picture of Nitrogen bases – adenine, guanine, cytosine, thymine, uracil.

Diagnostic features:

1. Adenine:

- Adenine is a Purine nitrogenous base found in both DNA and RNA.
- In DNA, adenine binds with thymine. In RNA, adenine binds with uracil.

2. Guanine:

- Guanine is a Purine nitrogenous base found in both DNA and RNA.
- Both DNA and RNA, guanine paired with cytosine via three hydrogen bonds.

3. Cytosine:

- Cytosine is a Pyrimidine nitrogenous base found in both DNA and RNA.
- Both DNA and RNA, cytosine binds with guanine via three hydrogen bonds.

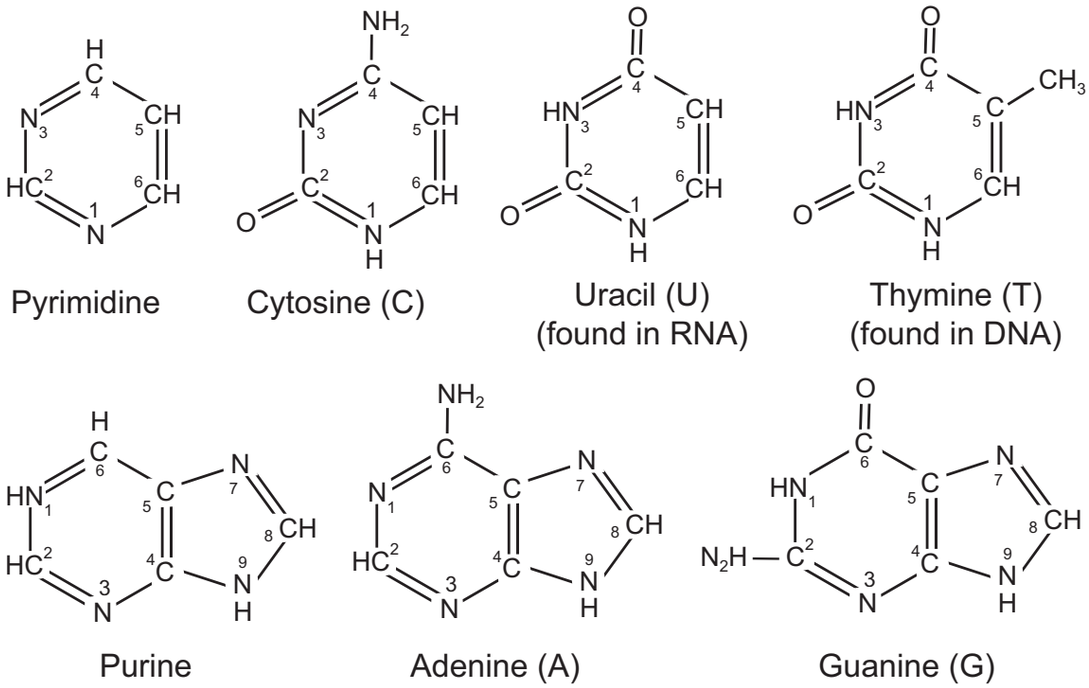


Figure 21: Nitrogen bases.

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